

Interactively Assisting Glaucoma Diagnosis with an Expert Knowledge-Distilled Vision Transformer

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Abstract

Optical Coherence Tomography (OCT) is critical for diagnosing eye diseases such as glaucoma. To improve the proficiency with which ophthalmologists-in-training diagnose glaucoma from OCT reports, we are developing a system that automatically identifies and highlights Regions of Interest (RoIs) in OCT reports. We accomplish this by distilling expert ophthalmologists' knowledge in a Vision Transformer (ViT) with their eye-tracking data. Our system features a graphical user interface that visually enhances AI-predicted RoIs via static and dynamic feedback modes, to help users cultivate systematic OCT-report viewing behaviors. Although no statistically significant improvement in diagnostic accuracy/confidence was observed with AI guidance, preliminary results indicate improved accuracy/confidence of ophthalmologists-in-training using RoIs identified by the expert-informed ViT-based approach compared to a conventional CNN-based approach. When using ViT-based guidance, participants exhibited less variance in viewing behavior compared to when they had no guidance, without compromising diagnostic accuracy. Furthermore, dynamic feedback yielded higher accuracy than static feedback.

CCS Concepts

• **Applied computing** → **Health informatics**; • **Human-centered computing** → **Interaction paradigms**; **User interface design**; **Scientific visualization**.

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719719>

Keywords

Computer-Aided Diagnosis, Vision Transformer, Optical coherence tomography

ACM Reference Format:

Ziheng Li, Haowen Wei, Kuang Sun, Leyi Cui, David Li, Steven Feiner, and Kaveri Thakoor. 2025. Interactively Assisting Glaucoma Diagnosis with an Expert Knowledge-Distilled Vision Transformer. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3719719>

1 Introduction

Glaucoma constitutes a significant challenge in global public health, accounting for 15% of the total burden of world blindness [36]. Optical Coherence Tomography (OCT) has emerged as a crucial imaging technique for diagnosing glaucoma and many other eye diseases [19, 37]. With the expected rise in the prevalence of eye disease due to an aging population, leveraging Artificial Intelligence (AI) to streamline diagnosis and minimize the demand on ophthalmology human resources is essential [29]. Previous studies have underscored the potential of computational intelligence to enhance clinical decision-making processes [25]. A majority of these studies have focused on training Machine Learning (ML) models to identify Regions of Interest (RoIs) in medical imaging [15, 18]. We introduce a novel approach for automatically suggesting RoIs during classification. Our system enables ophthalmologists-in-training to interactively learn from a model that harnesses distilled expert knowledge, derived from senior clinicians' eye movements on OCT reports during model training. This approach facilitates integrating expert-informed Computer-Aided Diagnosis (CAD) tools into routine clinical practice, thus bridging the gap between CAD and its practical application in ophthalmology. Our work represents a significant step toward integrating AI into clinical education and practice. The proposed system closes the loop: not only is the model trained using expert knowledge, but the trained model also passes down the expert knowledge to junior residents through an interactive graphical user interface that highlights RoIs. We thus make the

following contributions: (1) We propose expert-attention alignment loss—a methodology to train a Vision Transformer (ViT) model informed by experts’ eye-tracking data (going beyond previous work [22]); (2) We study the clinical relevance of various RoI-detection models as CAD tools; (3) We propose perceptual attention as a way to integrate auxiliary ViT outputs and the user’s attention to make possible personalized guidance when analyzing RoIs for OCT reports; (4) We present the first interactive graphical user interface and expert system that cues ophthalmologists to salient features in OCT reports, and conduct two user studies to evaluate our backend model and interactive system.

2 Related Work

2.1 Vision-based computer-aided diagnosis

Computer vision has significantly impacted medical processes involving imaging technologies, providing aids to tasks such as surgical planning and disease diagnosis [34, 41]. Models have been proposed to improve interaction with vision-based CAD systems by integrating Large Language Models (LLMs) to provide a supplementary natural language interface [38]. CAD tools in ophthalmology prioritize automatic disease detection [16, 28, 42]. Interactive CAD tools for ophthalmic imaging, such as fundus and OCT images, remain limited to targeting classification [10, 26] and segmentation [11]. Nevertheless, advanced automatic screening tools provide clinicians with minimal explanation of the decision-making process used by the models, limiting their clinical utility [23]. Kim et al. [24] attempt to address the gap in interactive tools with a web application that supports uploading fundus images for glaucoma classification. This system employs gradient-weighted class activation mapping (Grad-CAM) to visualize the diseased area, offering a more intuitive understanding of the diagnostic process. Additionally, Calisto et al. [5] highlight the importance of human-centered AI assistants that improve explainability and radiologist trust in AI-driven medical imaging. Their study demonstrates how interactive AI-generated heatmaps can improve diagnostic accuracy and cognitive workload.

2.2 Deep learning identifying pathological RoIs

Deep learning methods have recently come to center stage for automatic identification of RoIs in medical images [2, 12, 33]. This capability is not only essential for accurate classification but also for providing insights into disease-specific areas, thereby aiding clinicians in making informed decisions [8, 21, 25]. Studies have demonstrated the use of deep learning for detecting and analyzing RoIs in OCT images [9, 14, 17, 24, 39]. Transformers, which have recently gained traction in medical imaging, have been explored for their efficacy in RoI identification [15, 18]. As the visual analog of natural-language-processing transformers, ViTs are based on two fundamental principles: (1) positional embeddings, which describe the ordering of image patches (top-to-bottom, left-to-right, similar to the ordering of words in natural language) and (2) self-attention, whereby the model learns relative context (strength of association) between image patches (similar to certain words having a stronger likelihood of appearing together in language). While previous models explicitly predict the RoI, attention rollout is a

method to implicitly extract discriminating regions from transformers by taking the dot product between the queries and the keys [1, 7]. Attention rollout is similar to CAM in that no RoI information is provided during the training process, and the ability to locate relevant features is derived naturally from other training objectives such as classification [6]. Mbakwe et al. [27] conducted a qualitative analysis using attention rollout on chest scans, demonstrating the capability of their model to focus on pertinent regions. Despite the promising results of automatic RoI detection in medical images, more studies are needed to translate ML-based CAD systems into clinically-validated protocols, and make them understandable to practitioners [23]. Ibragimov and Mello-Thoms [20] reviewed the challenges of integrating AI-generated attention maps into clinical workflows, emphasizing eye-tracking-informed AI models as a potential solution to improve usability and trust in AI-assisted diagnosis. They highlighted that while AI-generated attention maps provide insights into disease localization, their reliability in real-world medical settings requires further validation. Salas et al. [32] designed a study in which clinicians were asked to classify OCT reports, while being presented with the predicted probability of glaucoma and a Grad-CAM from a Convolutional Neural Network (CNN). They found that clinicians reported increased confidence when the information from the model was present. Their work represents initial steps toward bringing ML-based RoI identification for medical images out of the realm of qualitative observations to be tested in formal user studies. Beede et al. [3] conducted an assessment of human expectations and post-deployment experiences of a diabetic retinopathy detection deep learning system in 11 eye hospitals in Thailand. Ours is one of the first CHI studies carried out for interactive glaucoma detection using an AI-plus-eye-tracking system.

2.3 Eye-tracking-informed models for medical images

Several innovative approaches have been developed recently to utilize eye-tracking data for training models that process medical images. One such approach is the RadioTransformer [4], which employs an Intersection-with-Union loss function. This method designates a student network to learn from a teacher network trained on radiologists’ eye-tracking data, enhancing the model’s ability to identify RoIs in chest scans. Patra et al. [30] present a sonograph model, another example of a teacher–student knowledge-transfer framework. This model benefits from sonographer gaze data to pinpoint RoIs and achieves comparable results to models with many more parameters. Stember et al. [35] utilize segmentation masks generated from eye-tracking data to train models that predict semantic segmentation of radiology images; they find that a model trained on eye-tracking masks is comparable to one trained with hand annotation. In the realm of OCT images, Kaushal et al. [22] present the Ophthalmologist-Gaze-Augmented ViT, which adds a term to the ViT loss function that contrasts ViT self-attention with ophthalmic-resident eye-gaze data. Our ViT guidance extends this approach.

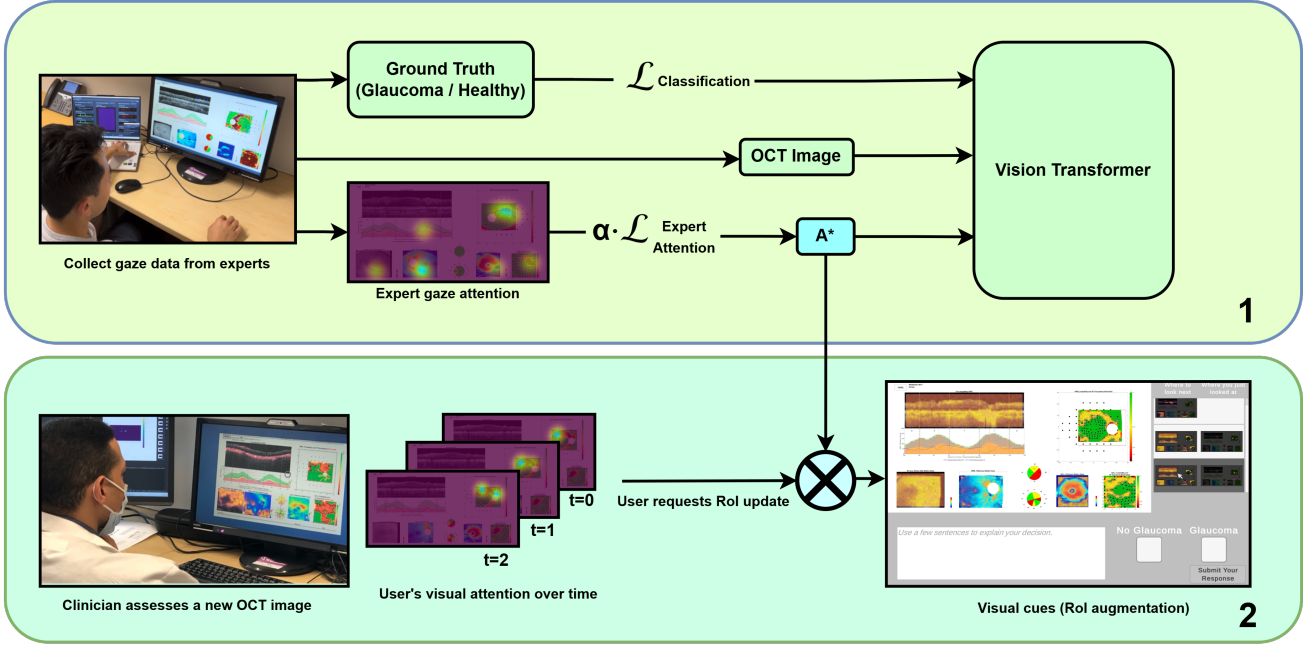


Figure 1: Our system aids ophthalmologists in diagnosing eye diseases such as glaucoma using OCT reports. (1) We collect visual attention data from clinical experts as they examine OCT images, which is then used to enhance a ViT-based classifier. (2) When clinicians view new OCT reports, they receive dynamic guidance that highlights key RoIs, tailored both by the model’s predictions and the clinician’s real-time visual attention.

3 Methods: Static AI-Based RoI Guidance

Our system (shown in Figure 1) detects RoIs in OCT reports that are used to diagnose glaucoma. The two key components of our system are: (1) a ViT trained and distilled with experts’ knowledge, and (2) an interactive front end that lets clinicians explore the discriminating features produced by the model for reports not seen during the training process. The front end combines the RoIs detected by the model and the user’s attention as they analyze the reports to deliver personalized guidance. We first discuss the ViT- and CNN-based unsupervised RoI detection models we developed, of which the former is a relatively new approach: only CNN-based models have been quantitatively studied for their clinical impact [32]. Next, we present our dataset and how we augment the ViT model’s learned self-attention with experts’ eye-tracking data with a method that we term *expert-attention loss* in Section 3.1.

3.1 Distilling expert knowledge via expert-alignment loss

To prepare a model that can highlight parts of an image that are more clinically relevant, particularly with relatively small medical image datasets, we augment ViT models with eye-tracking data from clinical experts. Effectively, we distill expert knowledge, embed it into a ViT, and provide this knowledge to novice participants through our user interface. Our OCT image dataset was collected from 185 eyes of 185 patients who visited the Columbia University Medical Center between 2010 and 2023 (IRB-approved study). This

patient population was inherently diverse due to the demographic composition (including minority groups of Hispanic, African American, and Asian descent) characteristic of Columbia’s location. We utilized this modest subset of data, as the ground-truth glaucoma diagnoses for these patients was generated from a consensus of 2–3 expert ophthalmologists who had access to OCT reports, visual field tests, and chart history for these patients (unlike our deep learning system, which is trained to detect glaucoma using OCT reports alone). Acquiring such a comprehensive ground-truth is a rare occurrence, explaining our relatively small dataset size. The ViT model is trained on *expert-attention loss* in addition to standard classification loss. The idea is that a certain amount of guidance from expert attention can improve classification performance and, in turn, the clinical relevance of a RoI generated by the model. How much expert-attention loss $\mathcal{L}_{ExpAttn}$ contributes to the training is controlled by the coefficient α in Eq. 1:

$$\begin{aligned} \mathcal{L}_{ExpAttn} &= \text{CrossEntropy}[\text{AttentionRollout}(L), A^*](1) \\ \mathcal{L} &= \mathcal{L}_{cls} + \alpha \times \mathcal{L}_{ExpAttn}, \end{aligned} \quad (2)$$

where $\mathcal{L}_{cls}$ is the binary cross entropy loss for the classification task and A^* is the expert attention heatmap (for directly contrasting with the ViT attention rollout). To investigate the effect of expert-attention loss and model size, we tested two different ViTs: Simple-ViT (with a single attention block) and Pretrained Small ViT (with twelve attention blocks and with pretrained weights from HuggingFace [13]). Interested readers can see Appendix Table 1

for details on how different models perform and Appendix Table 2 for architectural details, as well as technical definition of Attention Rollout.

3.2 User studies to evaluate expert-knowledge distilled AI guidance

We aim to find the model that generates RoIs with the most clinical relevance and improves clinicians' diagnostic performance. Through two user studies, we test the following hypotheses: (H1) The RoIs generated from attention rollout of a ViT model have higher clinical relevance than those generated via Grad-CAM from CNN-based models; (H2) The proposed loss function incorporating expert attention (Eq. 1) improves glaucoma classification performance (e.g., F1 score) while maintaining parameter efficiency; (H3) Clinical performance is improved when the user can interact with the perceptual RoI (described in the next section) while analyzing the OCT reports. H2 will be tested by evaluating the model's glaucoma classification performance (e.g., F1 score) and its parameter efficiency, as illustrated in Appendix Tables 1 and 2. We use classification accuracy as a proxy for the quality of the RoIs generated by the model. To test H1, we pick one CNN-based model and one ViT model with the highest classification performances to produce RoIs for the clinicians to view. In addition to the static heatmap derived from the model's RoI, we introduce perceptual RoI cuing, where the user can request updates to the RoI while actively viewing an OCT report, as described below.

4 Methods: Dynamic AI-Based RoI Guidance/Perceptual RoI Cuing

While methods such as attention rollout and Grad-CAM offer a glimpse into a model's decision criterion, a model's RoI may not always be interpretable by a clinician, especially given the intricacy of medical images, wherein subtle features can significantly influence a clinical decision. We propose *perceptual RoI cuing*, whereby the user can explore a model's attention in an intuitive way. This is in contrast to displaying the RoI straight from the model, which we call *static RoI*. The aim is to (1) take advantage of the self-attention between image patches, which does not use regular attention rollout techniques [7], and (2) foster more personalized collaboration between the clinician and the ML backend by letting them explore the model's knowledge combined with their own experience. Specifically, we formulate an algorithm to integrate the user's attention at the time of viewing the image and the attention rollout from the ViT. When the user requests an update to the RoIs, the newly computed RoIs will be the regions that are most informative for classification given the user's attention so far. We formalize this claim through a proof sketch in the Appendix.

5 User Studies

Our research employs two user studies to evaluate the efficacy of different RoI guidance methods in clinical decision-making for OCT images in ophthalmology. There are two variables under scrutiny: First, we investigate the model used to produce the RoI heatmap. To compare the clinical relevance of rollout from ViT and that of the Grad-CAM from CNN [32] among the CNN-based models we tested, we use the classification performance as a heuristic for the quality

of their Grad-CAM. In this case, *ResNet50* performs the best with an averaged cross-fold accuracy of 94.7%. Second, we investigate whether being able to personalize the RoI with the perceptual attention (Section 4) gives an edge over the "static" heatmap generated by the model without input from the user. Note that the perceptual RoI can only be computed from a ViT. Combining the variables gives us three conditions for User Study 1: (1) No RoI guidance; Static RoI guidance using a heatmap generated from (2) either *Grad-CAM* of the best-performing CNN-based model (*ResNet50*) or (3) the best-performing Vision Transformer. (We abstracted the details of the backend for participants, calling them *Model A* (ViT+Rollout) and *Model B* (GradCAM+ResNet) in the interface.) Thus, User Study 1 tests *H1* and *H2*. User Study 2 has two conditions: (1) Static RoI guidance using a heatmap generated from the ViT+Rollout; (2) Interactive RoI with perceptual attention, integrating the user's real-time attention and the model's attention (as detailed in Section 4). Thus, User Study 2 tests *H3*.

5.1 Experiment protocol

We recruited eight participants: five ophthalmology residents, one OCT technician, and two experts (glaucoma specialists) for Study 1. For Study 2, we recruited five different participants: two ophthalmology residents, one OCT technician, and two experts (glaucoma specialists). The studies use a Tobii Pro Fusion 250 Hz eye-tracker mounted on the bezel of a 1920 × 1080-pixel monitor with a 60-Hz refresh rate on which the OCT reports were displayed. Participants sat comfortably in front of the monitor, with access to a keyboard and mouse to interact with the paradigm.

Each study started with eye-tracker calibration for the participant. Each study proceeded in a series of blocks, one for each condition, each starting with a practice trial, followed by a fixed number of timed trials, as described below. We counterbalanced condition order across participants. Instructions were provided before each block to inform participants of the upcoming condition. In each trial, participants were required to determine whether glaucoma is present, type a brief rationale (a few sentences) for the decision, and rate their confidence on a four-point scale, from 0 ("Limited Confidence") to 3 ("Very Confident"). Each session was designed to last less than 30 minutes. Upon completion of the trials, participants were given a post-experiment survey to gather feedback on their experience. In between each trial, we briefly checked eye-tracker calibration and fixed it if necessary. For study 1, the timed trials in each block consisted of three glaucomatous (G) and three healthy (H) images; their order was randomized within each block. The three counterbalanced blocks were No Guidance, GradCAM+ResNet, and ViT+Rollout. For study 2, five G and five H images were shown with static (ViT+Rollout) guidance, and five G and five H images were shown for perceptual guidance.

5.2 Participant interaction

In all guidance conditions (GradCAM+ResNet, ViT+Rollout), interaction options included toggling guidance overlay visibility and adjusting relative brightness between the report and the guidance. In the perceptual guidance condition, in addition to highlighting RoIs, participants could request dynamic guidance updates based on their real-time visual attention from eyetracking and access the

previously generated guidance from a scroll area on the right of the screen (Figure 2d). These interactions allow participants to receive dynamic guidance based on their own visual attention, see their own visual attention, and navigate their history. Figure 2 illustrates all four conditions in our two user studies.

6 Results and Discussion

6.1 Quantitative analysis of glaucoma diagnosis accuracy and confidence

User Study 1. Table 1 compares no guidance, ViT-generated guidance, and CNN-generated guidance. Both the no-guidance and ViT-guidance scenarios demonstrated a comparable accuracy rate of 89.58%, while the CNN-guidance condition lagged slightly at 87.50%. Regarding confidence levels, participants felt marginally more confident with no guidance (1.73) compared to ViT (1.54) and CNN (1.46) guidance. On the other hand, qualitatively, five out of eight participants preferred ViT+Rollout (Model A) guidance, two out of eight found both GradCAM+ResNet (Model B) guidance and Model A guidance to be equivalent, and one out of eight preferred GradCAM+ResNet (Model B) guidance. This suggests that although there may not be a statistically significant improvement in accuracy or confidence across participants, the majority of users showed preference for the ViT+Rollout (Model A) guidance. We discuss some potential avenues to address the lack of significant accuracy or confidence improvement with Model A in Section 7. The qualitative preference for Model A (ViT+Rollout) justified our usage, as hypothesized in H1, of Model A as the first condition in User Study 2.

User Study 2. Table 1 also compares static guidance and perceptual guidance generated by ViT. The accuracy achieved under static guidance was 88%, while perceptual guidance led to a slightly higher accuracy of 90%. This improvement suggests that perceptual guidance, perhaps by offering more dynamic or context-sensitive cues, can enhance task performance slightly more effectively than static guidance. In terms of confidence, participants reported higher confidence with static guidance (2.18) compared to perceptual guidance (1.96). This indicates that while perceptual guidance may lead to better performance outcomes, static guidance seems to instill greater confidence in participants. This higher accuracy coupled with lower confidence could be attributed to the higher cognitive load incurred by the user when utilizing the perceptual mode compared to the static mode. This also aligns with prior work noting that lack of transparency can hurt user trust/confidence (i.e., when users do not know how the model generates its guidance) [31].

6.2 Correspondence between participant visual attention and RoI predictions

Method. To assess the correspondence between the RoIs suggested by our ViT and CNN-based models and the fixations made by participants, we computed the divergence (or relative entropy) of participant fixation distributions and model-predicted RoI distributions. We also carried out analysis of participant text responses and dwell time on OCT reports as a function of AI guidance; these results are

presented in the Appendix. The correspondence between participant visual attention and RoI predictions is presented in Figure 3.

Results. We found that the mean divergence for the no-guidance setting and ViT-based guidance setting (Figure 3a) was comparable, suggesting that our ViT-based model effectively learned RoIs that were comparable to the patterns of viewing behavior our participants inherently used/were taught to use (given the finding from Stember et al. [35] that hand-annotated RoIs are not significantly different from eye-tracked RoIs, this also validates that the AI has learned RoIs consistent with clinical expertise). However, interestingly, when using the ViT-based guidance, participants tended to exhibit less variance (shorter box-plot in Figure 3a) in that viewing behavior compared to when they had no guidance. This suggests that our ViT-based guidance was effective in improving participant efficiency of glaucoma diagnosis even though it was not effective at significantly improving glaucoma diagnosis accuracy or confidence compared to the no-guidance setting. Furthermore, as also indicated by higher qualitative preference for the ViT+Rollout over the GradCAM+ResNet condition in User Study 1, the CNN-based model did not support effective viewing behavior, as clinicians did not align with this model's predicted RoIs; Figure 3a exhibits that participants had very high relative entropy for the CNN-based guidance compared to that for ViT-based guidance or no-guidance.

6.3 Qualitative analysis of post-experiment survey

Following the completion of the experimental tasks, participants were asked to provide feedback on their experience through a post-experiment survey. Here, we present a thematic analysis of the responses, highlighting significant perceptions and experiences that emerged from the feedback.

User study 1: ViT vs. CNN guidance. The survey gathered user opinions on the system's interface, usability, and the effectiveness of ViT and CNN-generated guidance for identifying glaucoma cases. Participants found the interface intuitive and user-friendly. One described it as "intuitive, pretty easy to use, with the toggle to hide/show." (P4) Another noted its value for less experienced users: "Very helpful, especially for residents that have not had much exposure reading glaucoma images." (P9) ViT guidance was generally preferred over CNN, with participants highlighting its ability to pinpoint pathology more effectively. "I liked model A better because I feel like it pointed better the possible pathology." (P9) Another remarked, "Model A (ViT) looks at the more relevant parts of the RNFL and GCL maps." (P15) These responses suggest ViT's targeted attention rollout may better guide users to key regions compared to CNN's Grad-CAM approach.

User study 2: static ViT vs. interactive ViT. This study compared static and perceptual ViT-generated guidance in glaucoma diagnosis. It explored how different guidance modes impact diagnostic processes and user experience. Participants responded positively to the system's usability, describing it as "Easy to use, easy to understand." (P14) and "reasonably easy to use." (P15). Static guidance was appreciated for its clarity and consistency, while perceptual guidance, which updates based on user interaction, was valued for directing attention to less obvious areas. However, some

	No Guidance	ViT Guidance	CNN Guidance	Static Guidance	Perceptual Guidance
Accuracy	89.58% $\pm$ 8.63%	89.58% $\pm$ 12.40%	87.50% $\pm$ 11.79%	88% $\pm$ 17.89%	90% $\pm$ 7.07%
Confidence	1.73 $\pm$ 1.04	1.54 $\pm$ 1.01	1.46 $\pm$ 0.64	2.18 $\pm$ 0.56	1.96 $\pm$ 0.39

Table 1: Study 1 Accuracy (left), Study 2 Accuracy (right)

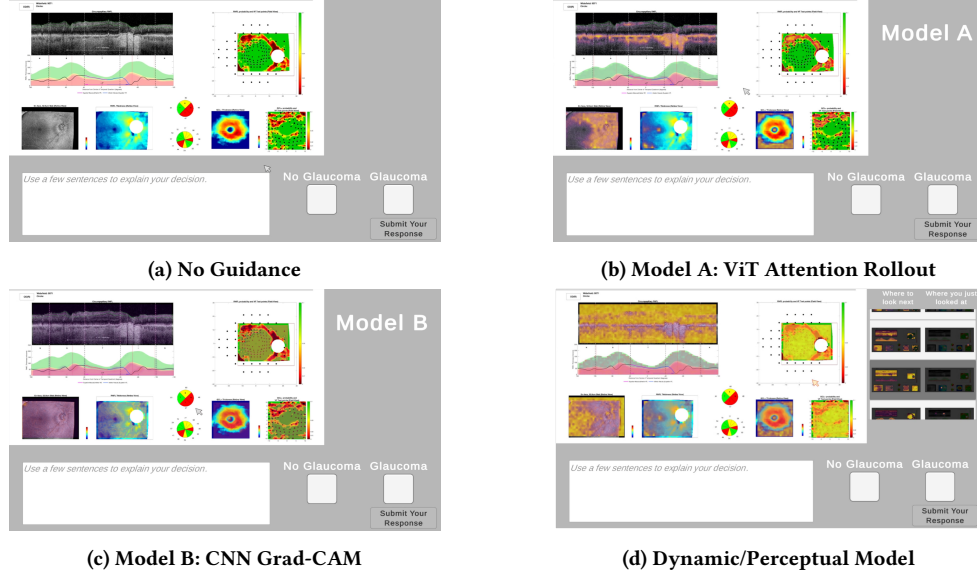
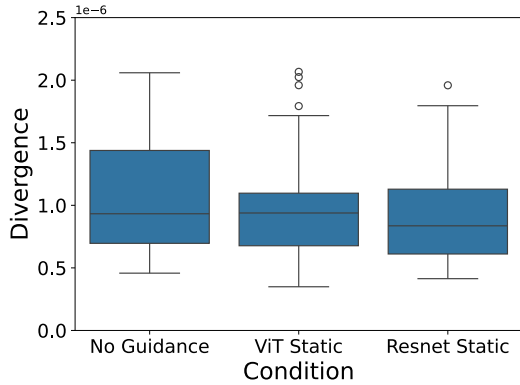


Figure 2: An example of the four conditions presented across our two user studies: (a) No guidance condition, (b) Model A: ViT+Rollout condition, (c) Model B: GradCAM+ResNet condition, (d) perceptual/dynamic model condition.



(a) Participant Gaze Divergence vs. Feedback Condition

$$relative\ entropy = \begin{cases} \log \frac{x}{y}, & x > 0, y > 0 \\ 0, & x = 0, y \geq 0 \\ \infty, & otherwise \end{cases} \quad (3)$$

(b) Definition of Divergence/Relative Entropy

Figure 3: (a) Median and variance for the divergence between participants' fixation distributions on the OCT image and models' predicted-ROI distributions. The ViT condition exhibits less variance than the no-guidance or ResNet conditions. (b) Definition of divergence (or relative entropy).

found it distracting when updates highlighted overly large image sections. Moving forward, this feedback suggests the need to balance interactive engagement with the clarity of static guidance. Enhancements will focus on improving intuitiveness and ensuring updates aid rather than complicate diagnostics.

7 Limitations and Future Directions

Given the cutting-edge nature of this work, our study has limitations that we aim to address in future work. One of the biggest limitations of our work currently is in User Study 2: As elaborated

in the qualitative feedback of some subjects, the interactive feedback would be more valuable to users if it were more localized to RoIs of the highest importance rather than showing multiple global regions, which may become obtrusive, increasing the cognitive load for the user. A well-designed system should subtly enhance the decision-making process [40]. In future work, we aim to address this by implementing a temporal CNN or transformer-based model to predict the next few patches of the highest importance conditioned on the previous patches viewed by the user rather than using the current Bayesian approach, which simply highlights opposing regions compared to what the user has already viewed. While we intend to increase our OCT-report dataset size and number of user-study participants in future work, for this pilot, late-breaking-work paper, this number is justified based on the limited availability of expert participants and well-labeled medical data. Moreover, adding more transparent or context-rich AI explanations (e.g., textual justifications or comparisons between prototype predictions) could help clinicians verify guidance more easily, improving their confidence in the final diagnoses.

8 Conclusions

We were able to effectively show evidence to support two of our three hypotheses. First, we distilled expert knowledge into our ViT model through our expert-aligned loss, and expert-informed ViT outperforms both pretrained ViT and CNNs (F1-score of 0.968 at expert-contribution coefficient α of 0.01). In the user studies, this was exhibited by the similar mean divergence exhibited by participants when they experienced no guidance or ViT static guidance in our user studies. Second, we observed that ViT-based guidance supported effective glaucoma diagnosis viewing behavior compared to CNN-based guidance. This was exhibited through marginally higher accuracy and confidence value achieved by participants when they received ViT guidance compared to when they received CNN-based guidance cues and preference by the majority of participants for the ViT guidance. In contrast, our current findings show that our user interface does not significantly enhance participants' clinical performance when using guidance cues from either model (regardless of the back-end model). Finally, we observed that the perceptual/interactive guidance slightly improved performance (accuracy of glaucoma diagnosis) compared to the static guidance, although it did not improve confidence in diagnoses. The more coarse features and sub-images present in OCT reports may have made our AI guidance redundant to what trainees are already taught, explaining the lack of significant improvements in accuracy and confidence with our system. In future work, we will investigate the efficacy of our approach in assisting the detection of eye diseases using more complex OCT B-scans (containing more fine-grained instead of coarse features than OCT reports), potentially increasing the impact of ViT-based guidance on diagnostic performance. Our system has the potential to aid in medical education or as a corroboration/second opinion for experts; user study feedback suggests our easy-to-use interface could be adopted into the workflow of an ophthalmic clinic.

Acknowledgments

The authors would like to thank the reviewers and area chairs for their insightful comments which helped us to significantly improve the manuscript. We also would like to gratefully acknowledge the expert ophthalmology participants who contributed their eye tracking data for our AI model training and the medical trainee participants who evaluated our system through the user study described in the paper. Finally, we acknowledge our funding sources: Columbia University's Data Science Institute Seed Fund, an Unrestricted Grant from Research to Prevent Blindness, New York, NY, USA, and a Center of Excellence grant from the Air Force Office of Scientific Research (FA9550-22-1-0337). This study, Protocol AAAU079, was approved by the Columbia University Irving Medical Center Institutional Review Board on 12/30/2024 and is in accordance with the tenets set forth by the Declaration of Helsinki. Informed consent was obtained from all study participants.

References

- [1] Samira Abnar and Willem Zuidema. 2020. Quantifying attention flow in transformers. *arXiv preprint arXiv:2005.00928* (2020).
- [2] Adekanmi Adeyinka Adegun, Serestina Viriri, and Roseline Oluwaseun Ogun-dokun. 2021. Deep learning approach for medical image analysis. *Computational Intelligence and Neuroscience* 2021 (2021), 1–9.
- [3] Emma Beede, Elizabeth Baylor, Fred Hersch, Anna Iurchenko, Lauren Wilcox, Paisan Ruamviboonsuk, and Laura M Vardoulakis. 2020. A human-centered evaluation of a deep learning system deployed in clinics for the detection of diabetic retinopathy. In *Proceedings of the 2020 CHI conference on human factors in computing systems*. 1–12.
- [4] Moinak Bhattacharya, Shubham Jain, and Prateek Prasanna. 2022. RadioTrans-former: a cascaded global-focal transformer for visual attention-guided disease classification. In *European Conference on Computer Vision*. Springer, 679–698.
- [5] Francisco Maria Calisto, Carlos Santiago, Nuno Nunes, and Jacinto C Nascimento. 2021. Introduction of human-centric AI assistant to aid radiologists for mul-timodal breast image classification. *International Journal of Human-Computer Studies* 150 (2021), 102607.
- [6] Hila Chefer, Shir Gur, and Lior Wolf. 2021. Generic attention-model explainability for interpreting bi-modal and encoder-decoder transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 397–406.
- [7] Hila Chefer, Shir Gur, and Lior Wolf. 2021. Transformer interpretability beyond attention visualization. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 782–791.
- [8] Xuxin Chen, Ximin Wang, Ke Zhang, Kar-Ming Fung, Theresa C Thai, Kathleen Moore, Robert S Mannel, Hong Liu, Bin Zheng, and Yuchen Qiu. 2022. Recent advances and clinical applications of deep learning in medical image analysis. *Medical Image Analysis* 79 (2022), 102444.
- [9] Xiaoming Chen, Ying Xue, Xiaoyan Wu, Yi Zhong, Huiying Rao, Heng Luo, and Zuquan Weng. 2023. Deep Learning-Based System for Disease Screening and Pathologic Region Detection From Optical Coherence Tomography Images. *Translational Vision Science & Technology* 12, 1 (2023), 29–29.
- [10] Mohamed Elsharkawy, Ahmed Sharafelddeen, Ahmed Soliman, Fahmi Khalifa, Mohammed Ghazal, Eman El-Daydamony, Ahmed Atwan, Harpal Singh Sandhu, and Ayman El-Baz. 2022. A novel computer-aided diagnostic system for early detection of diabetic retinopathy using 3D-OCT higher-order spatial appearance model. *Diagnostics* 12, 2 (2022), 461.
- [11] Ahmed ElTanboly, Marwa Ismail, Ahmed Shalaby, Andy Switala, Ayman El-Baz, Shlomit Schaal, Georgy Gimel'farb, and Magdi El-Azab. 2017. A computer-aided diagnostic system for detecting diabetic retinopathy in optical coherence tomography images. *Medical physics* 44, 3 (2017), 914–923.
- [12] Bradley J Erickson, Panagiotis Korfiatis, Zeynettin Akkus, and Timothy L Kline. 2017. Machine learning for medical imaging. *Radiographics* 37, 2 (2017), 505–515.
- [13] Hugging Face. 2023. timm_vit_small_patch32_224.augreg_in21k_ft_in1k. https://huggingface.co/timm/vit_small_patch32_224.augreg_in21k_ft_in1k. Accessed: 2024-08-28.
- [14] Leyuan Fang, Chong Wang, Shutao Li, Hossein Rabbani, Xiangdong Chen, and Zhimin Liu. 2019. Attention to lesion: Lesion-aware convolutional neural network for retinal optical coherence tomography image classification. *IEEE transactions on medical imaging* 38, 8 (2019), 1959–1970.
- [15] Yunhe Gao, Mu Zhou, and Dimitris N Metaxas. 2021. UINet: a hybrid trans-former architecture for medical image segmentation. In *Medical Image Computing and Computer Assisted Intervention—MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part III* 24. Springer,

- 61–71.
- [16] Yuki Hagiwara, Joel En Wei Koh, Jen Hong Tan, Sulatha V Bhandary, Augustinus Laude, Edward J Ciaccio, Louis Tong, and U Rajendra Acharya. 2018. Computer-aided diagnosis of glaucoma using fundus images: A review. *Computer methods and programs in biomedicine* 165 (2018), 1–12.
 - [17] Bilal Hassan, Shiyin Qin, Ramsha Ahmed, Taimur Hassan, Abdel Hakeem Taguri, Shahrukh Hashmi, and Naoufel Werghi. 2021. Deep learning based joint segmentation and characterization of multi-class retinal fluid lesions on OCT scans for clinical use in anti-VEGF therapy. *Computers in Biology and Medicine* 136 (2021), 104727.
 - [18] Md Shakhawat Hossain, Galib Muhammad Shahriar, MM Mahbul Syeed, Mohammad Faisal Uddin, Mahady Hasan, Shingla Shivam, and Suresh Advani. 2023. Region of interest (ROI) selection using vision transformer for automatic analysis using whole slide images. *Scientific Reports* 13, 1 (2023), 11314.
 - [19] David Huang, Eric A Swanson, Charles P Lin, Joel S Schuman, William G Stinson, Warren Chang, Michael R Hee, Thomas Flotte, Kenton Gregory, Carmen A Puliafito, et al. 1991. Optical coherence tomography. *science* 254, 5035 (1991), 1178–1181.
 - [20] Bulat Ibragimov and Claudia Mello-Thoms. 2024. The Use of Machine Learning in Eye Tracking Studies in Medical Imaging: A Review. *IEEE journal of biomedical and health informatics* (2024).
 - [21] Huiyan Jiang, Zhaoshuo Diao, Tianyu Shi, Yang Zhou, Feiyu Wang, Wenrui Hu, Xiaolin Zhu, Shijie Luo, Guoyu Tong, and Yu-Dong Yao. 2023. A review of deep learning-based multiple-lesion recognition from medical images: classification, detection and segmentation. *Computers in Biology and Medicine* (2023), 106726.
 - [22] Shubham Kaushal, Yifan Sun, Ryan Zukerman, Royce WS Chen, and Kaveri A Thakoor. 2023. Detecting Eye Disease Using Vision Transformers Informed by Ophthalmology Resident Gaze Data. In *2023 45th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. IEEE, 1–4.
 - [23] Christopher J Kelly, Alan Karthikesalingam, Mustafa Suleyman, Greg Corrado, and Dominic King. 2019. Key challenges for delivering clinical impact with artificial intelligence. *BMC medicine* 17 (2019), 1–9.
 - [24] Mijung Kim, Jong Chul Han, Seung Hyup Hyun, Olivier Janssens, Sofie Van Hoecke, Changwon Kee, and Wesley De Neve. 2019. Medinoid: computer-aided diagnosis and localization of glaucoma using deep learning. *Applied Sciences* 9, 15 (2019), 3064.
 - [25] SV Mahesh Kumar and R Gunasundari. 2023. Computational intelligence in eye disease diagnosis: a comparative study. *Medical & Biological Engineering & Computing* 61, 3 (2023), 593–615.
 - [26] Yutao Ma, Tao Xu, Xiaolei Huang, Xiaofang Wang, Canyu Li, Jason Jerwick, Yuan Ning, Xianxu Zeng, Baojin Wang, Yihong Wang, et al. 2019. Computer-aided diagnosis of label-free 3-D optical coherence microscopy images of human cervical tissue. *IEEE Transactions on Biomedical Engineering* 66, 9 (2019), 2447–2456.
 - [27] Amarachi B Mbakwe, Lyuyang Wang, Mehdi Moradi, and Ismini Lourentzou. 2023. Hierarchical Vision Transformers for Disease Progression Detection in Chest X-Ray Images. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 685–695.
 - [28] Delaram Mirzania, Atalie C Thompson, and Kelly W Muir. 2021. Applications of deep learning in detection of glaucoma: a systematic review. *European Journal of Ophthalmology* 31, 4 (2021), 1618–1642.
 - [29] Raffaele Nuzzi, Giacomo Boscia, Paola Marolo, and Federico Ricardi. 2021. The impact of artificial intelligence and deep learning in eye diseases: a review. *Frontiers in Medicine* 8 (2021), 710329.
 - [30] Arijit Patra, Yifan Cai, Pierre Chatelain, Harshita Sharma, Lior Drukker, Aris T Papageorgiou, and J Alison Noble. 2019. Efficient ultrasound image analysis models with sonographer gaze assisted distillation. In *Medical Image Computing and Computer Assisted Intervention—MICCAI 2019: 22nd International Conference, Shenzhen, China, October 13–17, 2019, Proceedings, Part IV* 22. Springer, 394–402.
 - [31] Drew Prinster, Amama Mahmood, Suchi Sarria, Jean Jeudy, Cheng Ting Lin, Paul H Yi, and Chien-Ming Huang. 2024. Care to explain? AI explanation types differentially impact chest radiograph diagnostic performance and physician trust in AI. *Radiology* 313, 2 (2024), e233261.
 - [32] Joel Salas, Ryan Zukerman, Omar Moussa, Sophie Z Gu, Ari Leshno, Jeffrey M Liebmann, George Cioffi, and Kaveri Thakoor. 2023. Impact of AI on Retrospective Glaucoma Diagnosis. *Investigative Ophthalmology & Visual Science* 64, 8 (2023), 385–385.
 - [33] Dinggang Shen, Guorong Wu, and Heung-Il Suk. 2017. Deep learning in medical image analysis. *Annual review of biomedical engineering* 19 (2017), 221–248.
 - [34] Hyewon Song, Anh-Duc Nguyen, Myoungsik Gong, and Sanghoon Lee. 2016. A review of computer vision methods for purpose on computer-aided diagnosis. *Journal of International Society for Simulation Surgery* 3, 1 (2016), 1–8.
 - [35] Joseph N Stember, Haydar Celik, E Krupinski, Peter D Chang, Simukayi Mutasa, Bradford J Wood, A Lignelli, Gul Moonis, LH Schwartz, Sachin Jambawalikar, et al. 2019. Eye tracking for deep learning segmentation using convolutional neural networks. *Journal of digital imaging* 32 (2019), 597–604.
 - [36] B Thylefors and AD2486713 Negrel. 1994. The global impact of glaucoma. *Bulletin of the World Health Organization* 72, 3 (1994), 323.
 - [37] Muhammad Usman, Muhammad Moazam Fraz, and Sarah A Barman. 2017. Computer vision techniques applied for diagnostic analysis of retinal OCT images: a review. *Archives of Computational Methods in Engineering* 24 (2017), 449–465.
 - [38] Sheng Wang, Zihao Zhao, Xi Ouyang, Qian Wang, and Dinggang Shen. 2023. Chatcad: Interactive computer-aided diagnosis on medical image using large language models. *arXiv preprint arXiv:2302.07257* (2023).
 - [39] Xi Wang, Hao Chen, An-Ran Ran, Luyang Luo, Poemen P Chan, Clement C Tham, Robert T Chang, Suria S Mannil, Carol Y Cheung, and Pheng-Ann Heng. 2020. Towards multi-center glaucoma OCT image screening with semi-supervised joint structure and function multi-task learning. *Medical Image Analysis* 63 (2020), 101695.
 - [40] Qian Yang, Aaron Steinfeld, and John Zimmerman. 2019. Unremarkable AI: Fitting intelligent decision support into critical, clinical decision-making processes. In *Proceedings of the 2019 CHI conference on human factors in computing systems*. 1–11.
 - [41] Xiao-Xia Yin, Sillas Hadjiloucas, and Yanchun Zhang. 2017. *Pattern classification of medical images: computer aided diagnosis*. Springer.
 - [42] Mohammad JM Zedan, Mohd Asyraf Zulkifley, Ahmad Asrul Ibrahim, Asraf Mohamed Moubark, Nor Azwan Mohamed Kamari, and Siti Raihanah Abdani. 2023. Automated glaucoma screening and diagnosis based on retinal fundus images using deep learning approaches: a comprehensive review. *Diagnostics* 13, 13 (2023), 2180.